Loan Default Risk Analysis

Summary of Findings & Recommendations

1. Project Overview

This analysis examines 601 personal loan applications (merged with 500 borrower profiles) to identify drivers of default and surface behavioural and geographic patterns useful for risk management.

Overall default rate in the portfolio: 24.3%.

Tools used: Python (pandas), matplotlib / seaborn. This is the author's first published data-analytics project.

2. Analysis Approach

- Loaded and validated both source files; confirmed no null values.
- Merged on borrower_id to create a single analysis table.
- Created categorical segments for credit score, DTI ratio and interest rate.
- Examined default patterns and Pearson correlations with numeric features.
- Extended the work with loan-purpose amount differences and state-level default rates (including sample sizes).

3. Key Correlations with Default

Feature	Correlation	Implication
Days Delinquent	+0.88	Very strong positive link. Loans past 30 days delinquent are high risk.
Credit Score	−0.29	Lower scores associated with higher default rates.
Interest Rate	+0.20	Higher rates linked to higher default risk. Non-defaulters receive cheaper rates on average.
DTI Ratio	+0.19	Higher debt-to-income increases likelihood of default.

Highest default counts appear in the 500–599 credit-score band and in the 51–75% DTI range. Defaults are also concentrated in the 9–12% and Above-12% interest-rate brackets.

4. Loan Purpose – Behavioural Differences

Average loan amounts differ between non-defaulters and defaulters:

Loan Purpose	Non-defaulters	Defaulters	Difference
Auto Loan	$18,405	$22,644	Non-defaulters borrow ~$4.2k less
Business Loan	$22,480	$16,407	Non-defaulters borrow ~$6.1k more
Education	$17,115	$29,183	Non-defaulters borrow ~$12.1k less
Debt Consolidation	$21,030	$29,036	Non-defaulters borrow ~$8.0k less
Medical Expenses	$21,298	$30,421	Non-defaulters borrow ~$9.1k less

Possible interpretations:

- Smaller Auto, Education and Debt Consolidation loans may be more sustainable for borrowers.
- Larger Business loans among non-defaulters may reflect businesses that generate cash flow supporting repayment — these relationships are worth monitoring after disbursement.

5. Geographic Insights (State Level)

Sample sizes are small for many states; default rates on small samples are volatile and should be treated as directional only.

State	Loans	Default Rate	Avg Income	Note
VA	12	58.3%	$57.5k	Highest rate (small sample)
CA	12	50.0%	$63.6k	High rate despite higher income
MO	31	41.9%	$58.1k	High rate, larger sample
TX	12	41.7%	$56.5k	High rate
CO	22	31.8%	$65.7k	Elevated rate with higher income
NC	33	18.2%	$67.8k	Lower rate, higher income
PA	12	16.7%	$56.8k	Same size as CA/TX but far lower default
LA	18	0.0%	$74.9k	No defaults in sample

Notable comparisons:

- PA vs CA (both 12 loans): 16.7% vs 50% default rate.
- NC vs MO (33 vs 31 loans): 18.2% vs 41.9% default rate. NC also shows ~$10k higher average income.
- Several lower-income states show elevated default rates (consistent with an income-related pattern), but CA and CO are exceptions that warrant further investigation.

6. Recommendations

- Tighten credit policy or add enhanced review for applicants with credit scores below 600 and DTI above 50%.
- Act early on any loan that reaches 30 days past due.
- Review underwriting and pricing for Home Improvement, Wedding, Education and larger Debt Consolidation loans.
- Investigate high-default states (CA, CO, VA, MO, TX) versus lower-default states (PA, NC, LA) for differences in borrower mix, product mix or local conditions.
- For Business loans, consider post-disbursement monitoring of business performance given the larger average amounts among non-defaulters.

7. Limitations

- Data covers only approximately two years (2024–2025) — insufficient for reliable trend analysis.
- Many states have small sample sizes (CA, PA, TX, VA each have only 12 loans). Rates on small samples are volatile.
- No collateral or home-value information is available for home-related loans.
- The analysis is descriptive and correlational; it does not establish causation.

8. Closing Note

This project was completed as a learning exercise in pandas, data visualisation and translating analysis into practical recommendations for a lending institution. Future iterations could add true default-rate calculations across all segments, simple risk flags that combine income + credit score + DTI, multi-year data, and a basic predictive model.